

IIT JAM 2014 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS

35

TOTAL MARKS

60 M

TEST DURATION

100 Min

PAPER PATTERN

MCQ (35 Questions)

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		35	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q1 • ID: JAM_2014_Q1)

MCQ

+1 M

Neg: -0.33

Q.1 Let $f(x) = |x^2 - 25|$ for all $x \in \mathbb{R}$. The total number of points of $\mathbb{R}$ at which f attains a local extremum (minimum or maximum) is

- (A) 1 (B) 2 (C) 3 (D) 4

Question 2 (JAM 2014 • Q2 • ID: JAM_2014_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 The coefficient of $(x-1)^2$ in the Taylor series expansion of $f(x) = xe^x$ ($x \in \mathbb{R}$) about the point $x=1$ is

- (A) $\frac{e}{2}$ (B) $2e$ (C) $\frac{3e}{2}$ (D) $3e$

Question 3 (JAM 2014 • Q3 • ID: JAM_2014_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let $f(x, y) = \sum_{k=1}^{10} (x^2 - y^2)^k$ for all $(x, y) \in \mathbb{R}^2$. Then for all $(x, y) \in \mathbb{R}^2$,

- (A) $x \frac{\partial f}{\partial x}(x, y) - y \frac{\partial f}{\partial y}(x, y) = 0$ (B) $x \frac{\partial f}{\partial x}(x, y) + y \frac{\partial f}{\partial y}(x, y) = 0$
(C) $y \frac{\partial f}{\partial x}(x, y) - x \frac{\partial f}{\partial y}(x, y) = 0$ (D) $y \frac{\partial f}{\partial x}(x, y) + x \frac{\partial f}{\partial y}(x, y) = 0$

Question 4 (JAM 2014 • Q4 • ID: JAM_2014_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 For $a, b, c \in \mathbb{R}$, if the differential equation

$$(ax^2 + bxy + y^2)dx + (2x^2 + cxy + y^2)dy = 0$$

is exact, then

- (A) $b = 2, c = 2a$ (B) $b = 4, c = 2$ (C) $b = 2, c = 4$ (D) $b = 2, a = 2c$

Question 5 (JAM 2014 • Q5 • ID: JAM_2014_Q5)

MCQ

+1 M

Neg: -0.33

- Q.5 If $f(x, y, z) = x^2y + y^2z + z^2x$ for all $(x, y, z) \in \mathbb{R}^3$ and $\nabla = \frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}$, then the value of $\nabla \cdot (\nabla \times \nabla f) + \nabla \cdot (\nabla f)$ at $(1, 1, 1)$ is
- (A) 0 (B) 3 (C) 6 (D) 9

Question 6 (JAM 2014 • Q6 • ID: JAM_2014_Q6)

MCQ

+1 M

Neg: -0.33

- Q.6 The radius of convergence of the power series $\sum_{n=0}^{\infty} 2^{2n} x^{n^2}$ is
- (A) $\frac{1}{4}$ (B) 1 (C) 2 (D) 4

Question 7 (JAM 2014 • Q7 • ID: JAM_2014_Q7)

MCQ

+1 M

Neg: -0.33

- Q.7 Let G be a group of order 17. The total number of non-isomorphic subgroups of G is
- (A) 1 (B) 2 (C) 3 (D) 17

Question 8 (JAM 2014 • Q8 • ID: JAM_2014_Q8)

MCQ

+1 M

Neg: -0.33

- Q.8 Which one of the following is a subspace of the vector space $\mathbb{R}^3$?
- (A) $\{(x, y, z) \in \mathbb{R}^3 : x + 2y = 0, 2x + 3z = 0\}$
- (B) $\{(x, y, z) \in \mathbb{R}^3 : 2x + 3y + 4z - 3 = 0, z = 0\}$
- (C) $\{(x, y, z) \in \mathbb{R}^3 : x \geq 0, y \geq 0\}$
- (D) $\{(x, y, z) \in \mathbb{R}^3 : x - 1 = 0, y = 0\}$

Question 9 (JAM 2014 • Q9 • ID: JAM_2014_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by $T(x, y, z) = (x + y, y + z, z + x)$ for all $(x, y, z) \in \mathbb{R}^3$. Then

(A) $\text{rank}(T) = 0$, $\text{nullity}(T) = 3$ (B) $\text{rank}(T) = 2$, $\text{nullity}(T) = 1$ (C) $\text{rank}(T) = 1$, $\text{nullity}(T) = 2$ (D) $\text{rank}(T) = 3$, $\text{nullity}(T) = 0$ **Question 10** (JAM 2014 • Q10 • ID: JAM_2014_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $x + \int_0^x f(t) dt = e^x - 1$ for all $x \in \mathbb{R}$. Then the set $\{x \in \mathbb{R} : 1 \leq f(x) \leq 2\}$ is the interval

(A) $[\log 2, \log 3]$ (B) $[2 \log 2, 3 \log 3]$ (C) $[e-1, e^2-1]$ (D) $[0, e^2]$ **Question 11** (JAM 2014 • Q11 • ID: JAM_2014_Q11)

MCQ

+2 M

Neg: -0.67

Q.11 The system of linear equations

$$x - y + 2z = b_1$$

$$x + 2y - z = b_2$$

$$2y - 2z = b_3$$

is inconsistent when (b_1, b_2, b_3) equals

(A) $(2, 2, 0)$ (B) $(0, 3, 2)$ (C) $(2, 2, 1)$ (D) $(2, -1, -2)$ **Question 12** (JAM 2014 • Q12 • ID: JAM_2014_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Let $A = \begin{bmatrix} a & -1 & 4 \\ 0 & b & 7 \\ 0 & 0 & 3 \end{bmatrix}$ be a matrix with real entries. If the sum and the product of all the

eigenvalues of A are 10 and 30 respectively, then $a^2 + b^2$ equals

(A) 29

(B) 40

(C) 58

(D) 65

Question 13 (JAM 2014 • Q13 • ID: JAM_2014_Q13)

MCQ

+2 M

Neg: -0.67

- Q.13 Consider the subspace $W = \{(x_1, x_2, \dots, x_{10}) \in \mathbb{R}^{10} : x_n = x_{n-1} + x_{n-2} \text{ for } 3 \leq n \leq 10\}$ of the vector space $\mathbb{R}^{10}$. The dimension of W is
- (A) 2 (B) 3 (C) 9 (D) 10

Question 14 (JAM 2014 • Q14 • ID: JAM_2014_Q14)

MCQ

+2 M

Neg: -0.67

- Q.14 Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions of the differential equation

$$x^2 y''(x) - 2xy'(x) - 4y(x) = 0 \text{ for } x \in [1, 10].$$

Consider the Wronskian $W(x) = y_1(x)y_2'(x) - y_2(x)y_1'(x)$. If $W(1) = 1$, then $W(3) - W(2)$ equals

- (A) 1 (B) 2 (C) 3 (D) 5

Question 15 (JAM 2014 • Q15 • ID: JAM_2014_Q15)

MCQ

+2 M

Neg: -0.67

- Q.15 The equation of the curve passing through the point $\left(\frac{\pi}{2}, 1\right)$ and having slope $\frac{\sin(x)}{x^2} - \frac{2y}{x}$ at each point (x, y) with $x \neq 0$ is

- (A) $-x^2 y + \cos(x) = \frac{-\pi^2}{4}$ (B) $x^2 y + \cos(x) = \frac{\pi^2}{4}$
(C) $x^2 y - \sin(x) = \frac{\pi^2}{4} - 1$ (D) $x^2 y + \sin(x) = \frac{\pi^2}{4} + 1$

Question 16 (JAM 2014 • Q16 • ID: JAM_2014_Q16)

MCQ

+2 M

Neg: -0.67

- Q.16 The value of $\alpha \in \mathbb{R}$ for which the curves $x^2 + \alpha y^2 = 1$ and $y = x^2$ intersect orthogonally is
- (A) -2 (B) $-\frac{1}{2}$ (C) $\frac{1}{2}$ (D) 2

Question 17 (JAM 2014 • Q17 • ID: JAM_2014_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 Let $x_n = 2^{2n} \left(1 - \cos \left(\frac{1}{2^n} \right) \right)$ for all $n \in \mathbb{N}$. Then the sequence $\{x_n\}$

(A) does NOT converge

(B) converges to 0

(C) converges to $\frac{1}{2}$ (D) converges to $\frac{1}{4}$

Question 18 (JAM 2014 • Q18 • ID: JAM_2014_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 Let $\{x_n\}$ be a sequence of real numbers such that $\lim_{n \rightarrow \infty} (x_{n+1} - x_n) = c$, where c is a positive real number. Then the sequence $\left\{ \frac{x_n}{n} \right\}$

(A) is NOT bounded

(B) is bounded but NOT convergent

(C) converges to c

(D) converges to 0

Question 19 (JAM 2014 • Q19 • ID: JAM_2014_Q19)

MCQ

+2 M

Neg: -0.67

Q.19 Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series, where $a_n = \frac{(-1)^n n}{2^n}$, $b_n = \frac{(-1)^n}{\log(n+1)}$ for all $n \in \mathbb{N}$. Then

(A) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are absolutely convergent(B) $\sum_{n=1}^{\infty} a_n$ is absolutely convergent but $\sum_{n=1}^{\infty} b_n$ is conditionally convergent(C) $\sum_{n=1}^{\infty} a_n$ is conditionally convergent but $\sum_{n=1}^{\infty} b_n$ is absolutely convergent(D) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are conditionally convergent

Question 20 (JAM 2014 • Q20 • ID: JAM_2014_Q20)

MCQ

+2 M

Neg: -0.67

Q.20 The set $\left\{ \frac{x^2}{1+x^2} : x \in \mathbb{R} \right\}$ is

- (A) connected but NOT compact in $\mathbb{R}$ (B) compact but NOT connected in $\mathbb{R}$
(C) compact and connected in $\mathbb{R}$ (D) neither compact nor connected in $\mathbb{R}$

Question 21 (JAM 2014 • Q21 • ID: JAM_2014_Q21)

MCQ

+2 M

Neg: -0.67

Q.21 The set of all limit points of the set $\left\{ \frac{2}{x+1} : x \in (-1, 1) \right\}$ in $\mathbb{R}$ is

- (A) $[1, \infty)$ (B) $(1, \infty)$ (C) $[-1, 1]$ (D) $[-1, \infty)$

Question 22 (JAM 2014 • Q22 • ID: JAM_2014_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 Let $S = [0, 1] \cup [2, 3)$ and let $f: S \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 2x & \text{if } x \in [0, 1], \\ 8-2x & \text{if } x \in [2, 3). \end{cases}$

If $T = \{ f(x) : x \in S \}$, then the inverse function $f^{-1}: T \rightarrow S$

- (A) does NOT exist (B) exists and is continuous
(C) exists and is NOT continuous (D) exists and is monotonic

Question 23 (JAM 2014 • Q23 • ID: JAM_2014_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 Let $f(x) = x^3 + x$ and $g(x) = x^3 - x$ for all $x \in \mathbb{R}$. If f^{-1} denotes the inverse function of f , then the derivative of the composite function $g \circ f^{-1}$ at the point 2 is

- (A) $\frac{2}{13}$ (B) $\frac{1}{2}$ (C) $\frac{11}{13}$ (D) $\frac{11}{4}$

Question 24 (JAM 2014 • Q24 • ID: JAM_2014_Q24)

MCQ

+2 M

Neg: -0.67

Q.24 For all $(x, y) \in \mathbb{R}^2$, let $f(x, y) = \begin{cases} x & \text{if } y = 0, \\ x - y^3 \sin(1/y) & \text{if } y \neq 0. \end{cases}$

Then at the point $(0, 0)$,

- (A) f is NOT continuous
- (B) f is continuous but NOT differentiable
- (C) $\frac{\partial f}{\partial x}$ exists but $\frac{\partial f}{\partial y}$ does NOT exist
- (D) f is differentiable

Question 25 (JAM 2014 • Q25 • ID: JAM_2014_Q25)

MCQ

+2 M

Neg: -0.67

Q.25 For all $(x, y) \in \mathbb{R}^2$, let $f(x, y) = \begin{cases} \frac{x}{|x|} \sqrt{x^2 + y^2} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$

Then $\frac{\partial f}{\partial x}(0, 0) + \frac{\partial f}{\partial y}(0, 0)$ equals

- (A) -1
- (B) 0
- (C) 1
- (D) 2

Question 26 (JAM 2014 • Q26 • ID: JAM_2014_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function with continuous derivative such that $f(\sqrt{2}) = 2$ and

$f(x) = \lim_{t \rightarrow 0} \frac{1}{2t} \int_{x-t}^{x+t} s f'(s) ds$ for all $x \in \mathbb{R}$. Then $f(3)$ equals

- (A) $\sqrt{3}$
- (B) $3\sqrt{2}$
- (C) $3\sqrt{3}$
- (D) 9

Question 27 (JAM 2014 • Q27 • ID: JAM_2014_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 The value of $\int_{x=0}^1 \int_{y=0}^{x^2} \int_{z=0}^y (y+2z) dz dy dx$ is

(A) $\frac{1}{53}$

(B) $\frac{2}{21}$

(C) $\frac{1}{6}$

(D) $\frac{5}{3}$

Question 28 (JAM 2014 • Q28 • ID: JAM_2014_Q28)

MCQ

+2 M

Neg: -0.67

Q.28 If C is a smooth curve in $\mathbb{R}^3$ from $(-1, 0, 1)$ to $(1, 1, -1)$, then the value of $\int_C (2xy + z^2) dx + (x^2 + z) dy + (y + 2xz) dz$ is

(A) 0

(B) 1

(C) 2

(D) 3

Question 29 (JAM 2014 • Q29 • ID: JAM_2014_Q29)

MCQ

+2 M

Neg: -0.67

Q.29 Let C be the boundary of the region $R = \{(x, y) \in \mathbb{R}^2 : -1 \leq y \leq 1, 0 \leq x \leq 1 - y^2\}$ oriented in the counterclockwise direction. Then the value of $\oint_C y dx + 2x dy$ is

(A) $-\frac{4}{3}$

(B) $-\frac{2}{3}$

(C) $\frac{2}{3}$

(D) $\frac{4}{3}$

Question 30 (JAM 2014 • Q30 • ID: JAM_2014_Q30)

MCQ

+2 M

Neg: -0.67

Q.30 Let G be a cyclic group of order 24. The total number of group isomorphisms of G onto itself is

(A) 7

(B) 8

(C) 17

(D) 24

Question 31 (JAM 2014 • Q31 • ID: JAM_2014_Q31)

MCQ

+2 M

Neg: -0.67

Q.31 Let S_n be the group of all permutations on the set $\{1, 2, \dots, n\}$ under the composition of mappings. For $n > 2$, if H is the smallest subgroup of S_n containing the transposition $(1, 2)$ and the cycle $(1, 2, \dots, n)$, then

- (A) $H = S_n$ (B) H is abelian
(C) the index of H in S_n is 2 (D) H is cyclic

Question 32 (JAM 2014 • Q32 • ID: JAM_2014_Q32)

MCQ

+2 M

Neg: -0.67

Q.32 Let S be the oriented surface $x^2 + y^2 + z^2 = 1$ with the unit normal $\mathbf{n}$ pointing outward. For the vector field $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, the value of $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$ is

- (A) $\frac{\pi}{3}$ (B) 2π (C) $\frac{4\pi}{3}$ (D) 4π

Question 33 (JAM 2014 • Q33 • ID: JAM_2014_Q33)

MCQ

+2 M

Neg: -0.67

Q.33 Let $f: (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x^2) = 1 - x^3$ for all $x > 0$ and $f(1) = 0$. Then $f(4)$ equals

- (A) $\frac{-47}{5}$ (B) $\frac{-47}{10}$ (C) $\frac{-16}{5}$ (D) $\frac{-8}{5}$

Question 34 (JAM 2014 • Q34 • ID: JAM_2014_Q34)

MCQ

+2 M

Neg: -0.67

Q.34 Which one of the following conditions on a group G implies that G is abelian?

- (A) The order of G is p^3 for some prime p
(B) Every proper subgroup of G is cyclic
(C) Every subgroup of G is normal in G
(D) The function $f: G \rightarrow G$, defined by $f(x) = x^{-1}$ for all $x \in G$, is a homomorphism

Question 35 (JAM 2014 • Q35 • ID: JAM_2014_Q35)

MCQ

+2 M

Neg: -0.67

Q.35 Let $S = \{x \in \mathbb{R} : x^6 - x^5 \leq 100\}$ and $T = \{x^2 - 2x : x \in (0, \infty)\}$. The set $S \cap T$ is

- (A) closed and bounded in $\mathbb{R}$
- (B) closed but NOT bounded in $\mathbb{R}$
- (C) bounded but NOT closed in $\mathbb{R}$
- (D) neither closed nor bounded in $\mathbb{R}$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2014 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+1	C
2	MCQ	+1	C
3	MCQ	+1	D
4	MCQ	+1	B
5	MCQ	+1	C
6	MCQ	+1	B
7	MCQ	+1	B
8	MCQ	+1	A
9	MCQ	+1	D
10	MCQ	+1	A
11	MCQ	+2	C
12	MCQ	+2	A

Q. No.	Type	Marks	Official Key
13	MCQ	+2	A
14	MCQ	+2	D
15	MCQ	+2	B
16	MCQ	+2	D
17	MCQ	+2	C
18	MCQ	+2	C
19	MCQ	+2	B
20	MCQ	+2	A
21	MCQ	+2	A
22	MCQ	+2	C
23	MCQ	+2	B
24	MCQ	+2	D

Q. No.	Type	Marks	Official Key
25	MCQ	+2	C
26	MCQ	+2	B
27	MCQ	+2	B
28	MCQ	+2	C
29	MCQ	+2	D
30	MCQ	+2	B
31	MCQ	+2	A
32	MCQ	+2	D
33	MCQ	+2	A
34	MCQ	+2	D
35	MCQ	+2	A

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.